ETALE LOCAL RINGS OF A NOETHERIAN ALGEBRAIC SPACE ARE NOETHERIAN LOCAL RINGS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 04KF: sources/stacks-chow/tags.txt, source lines 5971–5971. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06LJ: sources/stacks-chow/tags.txt, source lines 8468–8468. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\overline{x}$ be a geometric point of X. Then $\mathcal{O}_{X,\overline{x}}$ is a Noetherian local ring.*

Proof. Choose an étale neighbourhood $(U, \overline{u})$ of $\overline{x}$ where U is a scheme. Then $\mathcal{O}_{X,\overline{x}}$ is the strict henselization of the local ring of U at u, see Lemma 04KF. By our definition of Noetherian spaces the scheme U is locally Noetherian. Hence we conclude by More on Algebra, Lemma 06LJ. □

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.